
Foundation Models for Fast AC-State Evaluation: A GridSFM-Augmented Optimal Power Shutoff Diagnostic Study

Anonymous Author(s)

Affiliation

Address

email

Abstract

In this work, we embed the recently released GridSFM foundation model into the existing Optimal Power Shutoff problem as a testbed for foundation model transfer to a challenging topology-changing operating regime. On the GOC-500 bus grid example, we use GridSFM as a fast electrical-state predictor for externally generated line de-energization and load-service decisions and assess its behavior across wildfire-risk scenarios. We further fine-tune GridSFM using controlled variations in both training-set size and topology distribution to examine how adaptation affects predictive fidelity and downstream OPS performance. Fine-tuning improves state and physics metrics and produces closer agreement with fixed-decision AC-OPF reference solutions, while controlled warm-start experiments show only modest solver-speed benefits. Together, the observed speedups in candidate-state prediction and the evaluator-separated formulation establish a flexible framework for further studying foundation model integration in power-system decision-making.

1 Introduction

Optimal Power Shutoff and its computational challenge Wildfire mitigation creates a fundamental tradeoff for power-system operators: preventive line de-energization during Public Safety Power Shutoff (PSPS) events can reduce ignition exposure but may also interrupt customers and reduce served demand [4]. The Optimal Power Shutoff (OPS) problem formalizes this tradeoff as a topology-control problem in which high-risk lines are de-energized while continuous operational recourse, including generator redispatch, is adjusted to preserve power delivery. The original OPS formulation uses a linearized DC power-flow representation [11], while subsequent work shows that higher-fidelity AC formulations can more accurately represent network operation at substantially greater computational cost as system size and the number of candidate topologies increase [10]. For a fixed operating point, AC power flow requires solving the nonlinear active- and reactive-power balance equations, traditionally through iterative methods such as Newton–Raphson [12]. AC optimal power flow (AC-OPF) adds an optimization over operational controls while enforcing the nonlinear AC equations and associated operating constraints, and is commonly solved using nonlinear methods such as primal–dual interior-point algorithms [13]. In OPS, this difficulty is compounded by the combinatorial topology search: each candidate line-status decision creates a modified network whose operating state must be reevaluated.

Surrogates and foundation-model application Learned power-flow and optimal-power-flow surrogates offer one route around repeated nonlinear solves by directly predicting electrical states, including topology-aware models developed for contingency analysis and changing network configurations [3, 7]. Such models, however, are commonly specialized to a particular grid, task, or

operating distribution, motivating more reusable neural solvers and grid foundation models designed to transfer across heterogeneous systems and conditions [8, 14]. GridSFM, the released model studied here, is pretrained for AC-OPF across many grids and operating conditions and supports system-specific adaptation through fine-tuning [14]. Within wildfire-aware power-system applications, existing learning-based approaches have instead primarily targeted wildfire-risk estimation or the discrete line-de-energization decision [1, 6, 5, 2], rather than surrogate prediction of the resulting AC operating state. Accordingly, we use the OPS formulation of [11] as an operational reference and study whether a pretrained AC-OPF foundation model can provide fast state predictions for externally generated topology and load-service decisions. **To the best of our knowledge, this work is the first to integrate a pretrained grid foundation model into OPS as the surrogate AC-state predictor used to evaluate candidate topology and load-service decisions.**

2 GridSFM-Augmented OPS Framework

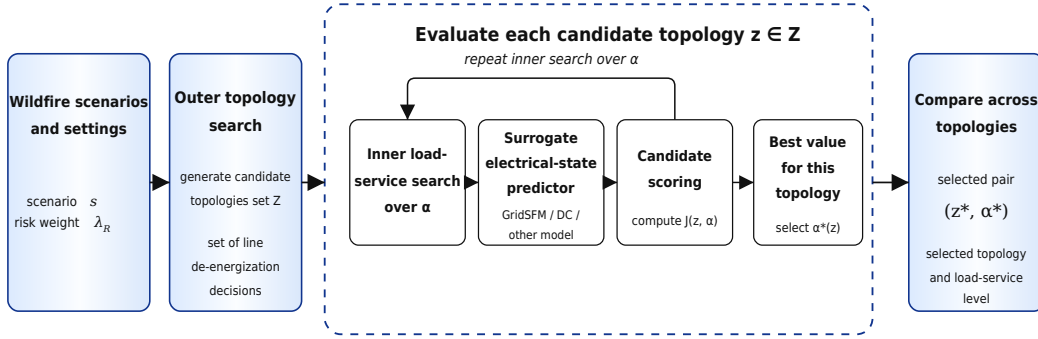


Figure 1: Overview of the framework which uses the existing grid data and wildfire risk scenario to create a **fast enumeration methodology**.

Outer topology search The outer master problem generates the set $\mathcal{Z}$ of promising line-de-energization candidates using low-cost proxies for wildfire exposure and load-service consequence. For a proxy risk weight λ_R^M , it solves

$$\begin{aligned} \min_z \quad & \lambda_R^M \hat{R}(z) + (1 - \lambda_R^M) \hat{L}(z) \\ \text{s.t.} \quad & \sum_{\ell \in \mathcal{L}} (1 - z_\ell) \leq K, \quad z_\ell \in \{0, 1\}, \quad K = 2, \end{aligned} \quad (1)$$

where $z_\ell = 1$ and $z_\ell = 0$ denote energized and de-energized lines, respectively. The proxy terms expose their dependence on the topology decision directly:

$$\hat{R}(z) = \frac{\sum_{\ell \in \mathcal{L}} w_\ell z_\ell}{\sum_{\ell \in \mathcal{L}} w_\ell}, \quad \hat{L}(z) = \sum_{\ell \in \mathcal{L}} c_\ell (1 - z_\ell). \quad (2)$$

Here, w_ℓ is a precomputed line-risk weight combining the scenario environmental wildfire exposure with the line's baseline loading, while c_ℓ is a precomputed service-impact score measuring the change in served demand associated with removing line ℓ from the baseline network.

GridSFM continuous recourse For each proposed topology $z \in \mathcal{Z}$, we optimize the load-service vector α over a reduced bounded space $\mathcal{A}_B(z)$ comprising five topology-relevant controls. Because GridSFM is treated as a black-box electrical evaluator, we use derivative-free Powell search over this reduced space, enabling computationally tractable evaluation across multiple candidate topologies. The best evaluated load-service decision $\alpha^*(z) \in \arg \min_{\alpha \in \mathcal{A}_B(z)} J(z, \alpha)$ is determined using the GridSFM-predicted AC operating state $\hat{x}_\theta = \text{GridSFM}(z, \alpha)$:

$$\begin{aligned} \min_{\alpha, \hat{x}_\theta} \quad & J(z, \alpha) = \lambda_R R_{\text{norm}}(z, \hat{x}_\theta) + (1 - \lambda_R) L_{\text{shed}}(\alpha) + \rho_{\text{phys}} P_{\text{AC}}^{\text{total}}(\hat{x}_\theta) \\ \text{s.t.} \quad & \hat{x}_\theta = \text{GridSFM}(z, \alpha), \\ & 0 \leq \alpha_i \leq 1, \quad i \in \mathcal{D}, \quad \alpha \in \mathcal{A}_B(z). \end{aligned} \quad (3)$$

61 The objective combines baseline-normalized wildfire exposure over the AC transmission-line set
 62 $\mathcal{L}_{\text{trans}}$, normalized unserved demand over the load-bus set $\mathcal{D}$, and a decomposed physics-infeasibility
 63 penalty:

$$R_{\text{norm}}(z, \hat{x}_\theta) = \frac{\sum_{\ell \in \mathcal{L}_{\text{trans}}} z_\ell p_\ell^{\text{env}} \text{loading}_\ell^2(z, \hat{x}_\theta)}{\sum_{\ell \in \mathcal{L}_{\text{trans}}} p_\ell^{\text{env}} (\text{loading}_\ell^{\text{base}})^2}, \quad L_{\text{shed}}(\alpha) = \frac{\sum_{i \in \mathcal{D}} P_{d,i}^{\text{base}} (1 - \alpha_i)}{\sum_{i \in \mathcal{D}} P_{d,i}^{\text{base}}}, \quad (4)$$

$$P_{\text{AC}}^{\text{total}} = P_{\text{AC}}^{\text{oper}} + P_{\text{AC}}^{P/Q} + P_{\text{AC}}^{\text{model}}.$$

64 Here, R_{norm} compares wildfire-weighted post-decision loading against the intact baseline, L_{shed}
 65 measures the fraction of baseline demand not served with α_i denoting the served fraction at load bus
 66 i , and $P_{\text{AC}}^{\text{total}}$ aggregates operational, active/reactive power-balance, and model-consistency constraint
 67 violations, where Table 1 provides a representative decomposition.

68 After retaining $\alpha^*(z)$ for each candidate topology, the final OPS solution is selected across the
 69 evaluated $(z, \alpha^*(z))$ pairs. Full formulation details, search settings, term definitions, islanding
 70 treatment, and differences from the original OPS formulation are provided in the Appendix.

71 3 Results

72 3.1 RQ1: GridSFM prediction performance in topology-changing OPS

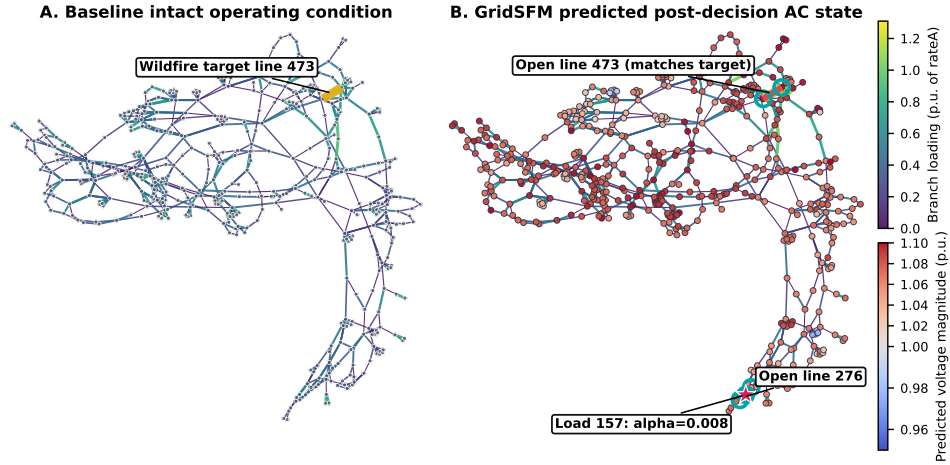


Figure 2: A depiction of the OPS workflow at use through the baseline operating condition and final OPS decision using the J-S1 wildfire scenario at $\lambda_R = 0.8$.

73 **Workflow and state response** We first evaluate GridSFM on the diagnostic J-S1 setting, which
 74 contains one high-wildfire-risk line with relatively low expected service impact and is designed to test
 75 whether OPS decisions follow the intended risk pattern. At $\lambda_R = 0.8$, the solution opens the targeted
 76 high-risk branch 473 and branch 276 and reduces load 157 to $\alpha_{\text{eff}} = 0.0081$. Predicted voltages
 77 remain within limits up to numerical tolerance. In this diagnostic setting, such interpretable state
 78 predictions provide a controlled proof of concept for using GridSFM inside OPS before considering
 79 larger, economically detailed wildfire-operation studies.

Physics consistency under topology shift Table 1 qualifies the preceding state prediction through explicit AC-physics constraint violations modeled in the objective function. Overall physics consistency degrades under the topology-changing operating regime, motivating the fine-tuning study in RQ2. Model-consistency violations remain zero across the evaluated states and serve primarily as a diagnostic for errors in the GridSFM evaluation workflow.

Table 1: Physics penalties for the J-S1 example.

Diagnostic	Intact	Modified
Voltage-limit penalty	0	0
Thermal-limit penalty	3.14×10^{-6}	1.58×10^{-4}
Generator-bound penalty	0	0
Source-less-service penalty	0	0
P_{AC}^{oper}	3.14×10^{-6}	1.58×10^{-4}
Active-power balance MSE	0.0768	0.0887
Reactive-power balance MSE	0.0250	0.0792
$P_{AC}^{P/Q}$	0.1018	0.1679
P_{AC}^{Total}	0.1018	0.1681

Candidate-state evaluation cost Across 54 unique fixed (z, α) instances, GridSFM achieves median $3.35\times$ and $3.59\times$ speedups over IPOPT for the core forward pass and complete candidate-to-state evaluation, respectively. Combined with the increased physics residuals under topology change, these results expose a clear speed-fidelity tradeoff: GridSFM substantially accelerates state evaluation, but improved physical fidelity remains necessary for reliable OPS assessment. Full timing distributions and paired confidence intervals are reported in the Appendix.

3.2 RQ2: Does fine-tuning improve GridSFM?

Fine-tuning performance Model 1 denotes the released GridSFM checkpoint. Model 2 fine-tunes Model 1 on 1,000 GOC-500 full topology cases; Model 3 continues Model 2 with 500 additional Full Topology samples, whereas Model 4 instead uses 500 N-1 samples.

Table 2: Fine-tuning evaluation and physics transfer.

Model	FullTop test		N-1 test		P_{AC}^{oper}		$P_{AC}^{P/Q}$	
	Loss	Br-P MAE	Loss	Br-P MAE	Intact	Modified	Intact	Modified
1	0.09330	0.07451	0.11847	0.07688	3.14×10^{-6}	1.58×10^{-4}	0.1018	0.1679
2	0.04079	0.04048	0.07015	0.04360	0	1.07×10^{-4}	0.0131	0.0738
3	0.03811	0.03502	0.06824	0.03869	0	1.09×10^{-4}	0.0080	0.0633
4	0.04038	0.03605	0.05680	0.03804	0	1.22×10^{-4}	0.0128	0.0312

Fine-tuning substantially improves both predictive performance and physics consistency relative to the released GridSFM checkpoint (Table 2). Although the topology-changing J-S1 regime still induces a structural shift between intact and modified states, all fine-tuned models substantially reduce the resulting AC-physics residuals. The lowest modified-state residual is obtained after adaptation with N-1 samples, suggesting that explicit exposure to topology variation is particularly useful for improving GridSFM behavior under OPS-induced network changes. These results motivate further study of topology-aware training/fine-tuning strategies for adapting foundation models to structurally shifted operating regimes. Full prediction metrics, residual constructions, and operational penalty definitions are provided in the Appendix.

3.3 RQ3: Do model improvements transfer to OPS behavior?

Native GridSFM tradeoffs. We compare Model 1–Model 4 within the common GridSFM evaluator, with all checkpoints sharing the same AC-loading definition, physics penalty, and search procedure. Fine-tuning shifts the empirical wildfire-risk/load-service tradeoffs toward lower native objective values (Figure 3). These fronts summarize the evaluated search budget rather than an exhaustive AC Pareto frontier.

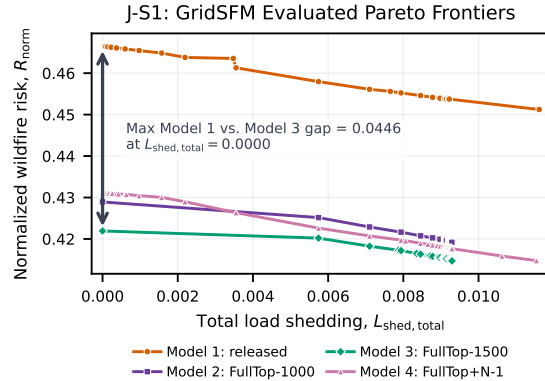


Figure 3: GridSFM Pareto tradeoffs on J-S1.

Transfer to the AC-OPF reference. More importantly, the gains observed during fine-tuning persist when selected OPS decisions are reevaluated against the common fixed-decision AC-OPF reference. From Model 1 to Model 3, state NRMSE decreases from 0.2073 to 0.1206, mean absolute wildfire-risk discrepancy from 0.0750 to 0.0176, and mean absolute objective discrepancy from 0.0308 to 0.0051, corresponding to approximately 42%, 76%, and 83% reductions. Thus, improved standalone GridSFM fidelity translates into substantially better downstream agreement with the higher-fidelity AC evaluation.

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Controlled warm starts. We finally test whether improved state prediction also accelerates nonlinear AC-OPF solution. Across the 75 fixed Reference A instances, median IPOPT iteration counts are 33 from a cold start, 30 from DC, and 31 for each GridSFM checkpoint. Median solve times are 5.113 s for cold initialization, 5.075 s for DC, and 5.104, 5.039, 5.055, and 5.067 s for Model 1–Model 4, respectively. Informed electrical states provide modest benefits over cold initialization, but improved GridSFM prediction accuracy does not induce a monotonic or proportional IPOPT speedup. Reference B provides a complementary diagnostic of whether the selected topology itself restricts AC-load deliverability; detailed results are deferred to the Appendix.

4 Conclusion and Outlook

Scope and findings This work provides a controlled diagnostic study of integrating a pretrained grid foundation model into topology-changing OPS as a surrogate AC-state predictor. On GOC-500, base GridSFM produces operationally interpretable candidate states while achieving a median $3.59\times$ speedup over fixed-decision AC-OPF at the complete candidate-to-state boundary. The topology-changing regime also increases physics residuals, motivating system-specific adaptation; subsequent fine-tuning improves predictive fidelity and substantially reduces disagreement with the fixed-decision AC-OPF reference. These improvements, however, do not translate proportionally into faster nonlinear-solver convergence. As the experiments use constructed wildfire scenarios and a bounded enumeration of topology and load-service candidates, the results establish a controlled proof of concept rather than a deployment-ready OPS policy.

Outlook The evaluator-separated formulation provides a basis for extending foundation-model-guided power-system decision making beyond this diagnostic setting. Security-constrained OPS

can test whether learned state prediction remains useful when shutoff decisions must preserve robustness to subsequent contingencies [9], while larger systems with realistic wildfire-risk data and explicit economic and reliability objectives can assess the operational value of the observed speed-fidelity tradeoff. Future work can also compare emerging grid foundation models under a common decision framework and move beyond external state prediction toward gray-box formulations in which foundation-model states and derivatives are coupled directly with the continuous optimization algorithm.

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184 A Additional RQ1 Timing Results

Table 3: Model 1 candidate-state evaluation timing over 54 unique fixed decisions. Times report median [IQR]; speedups are paired medians with 95% bootstrap intervals.

Boundary	GridSFM (s)	AC-OPF ref. (s)	Speedup
Core compute	0.222 [0.220–0.224]	0.746 [0.719–0.777]	$3.35\times$ [3.31–3.45]
State evaluator	0.297 [0.293–0.305]	1.092 [1.012–1.116]	$3.59\times$ [3.48–3.68]

185 B Full OPS formulation

- 186 • Decision variables, topology budget, effective load service, and source-less island rule.
- 187 • Load-shedding definition; wildfire-risk equation and intact AC normalization; PAC physics
- 188 penalty; full GridSFM candidate merit.
- 189 • Outer topology proxy and bounded α search.

190 C Evaluator and AC-reference definitions

- 191 • DC active-power loading versus GridSFM apparent-power loading and the native-objective
- 192 comparability boundary.
- 193 • Reference A fixed- (z, α) economic AC-OPF; Reference B maximum-load-delivery problem
- 194 and economic tie-break.
- 195 • Solver tolerances and interpretation of successful IPOPT solutions.

196 D Experimental and supplementary details

- 197 • J-S1/J-S2/J-S3 construction; λ_R sweep; $K \leq 2$; topology and continuous candidate budgets.
- 198 • Model 1–Model 4 fine-tuning splits, training settings, sealed full topology/N-1 metrics.
- 199 • RQ1 timing protocol, unique-instance provenance, full timing distributions, and paired
- 200 speedup statistics.
- 201 • All scenario Pareto plots; full Reference A/B results; full warm-start paired results.

202 AI4PowerGrids domain evaluation checklist

- 203 1. **Physics feasibility.** [FILL: State where full-AC feasibility / constraint-violation statistics
- 204 are reported (e.g., RQ1/RQ3 and appendix), or justify N/A.]
- 205 2. **Out-of-distribution evaluation.** [FILL: State where unseen topology / operating-
- 206 distribution evaluation is reported; distinguish FullTop and N-1 exposure carefully.]
- 207 3. **Failure modes.** [FILL: Point to worst-case/tail/qualitative physics or prediction failures,
- 208 or explicitly state the available failure-mode evidence and its limits.]
- 209 4. **System-level effect.** [FILL: Point to RQ3: native-to-reference OPS agreement, Pareto
- 210 behavior, and controlled warm-start study.]
- 211 5. **Data and code.** [FILL: Provide anonymous submission-time release plan or acceptance-
- 212 time release plan; state any restrictions.]